M13 Batch 7 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 563–650

Altitude alerting and reporting, clocks, flight data and voice recorders, condition monitoring, and gyroscopic principles.

232 questions · Level 3: 87 · Level 2: 104 · Level 1: 41

● marks a question built on a fact your instructor highlighted or underlined in the manual (175 of 232). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Altitude reporting and alerting

PDF pages 563–571 · 30 questions

LEVEL 1 — RECALL

1. The two warning conditions of the altitude alert system are ●
   a. Deviation warning and reset warning
   b. Approach warning and capture warning
   c. Approach warning and deviation warning

2. The altitude difference that activates the approach mode is ●
   a. 1000 ft
   b. 900 ft
   c. 300 ft

3. The modes of the digital altitude alert system are ●
   a. Approach, deviation, capture and warning
   b. Approach, deviation, alert and reset
   c. Approach, deviation, capture and reset

4. In the digital altitude alert system a deviation warning is given when ●
   a. the aircraft deviates 1000 ft from the selected altitude
   b. the aircraft deviates 900 ft from the selected altitude
   c. the aircraft deviates 300 ft from the selected altitude

LEVEL 2 — UNDERSTAND AND EXPLAIN

5. Where is the altitude alert system usually integrated ●
   a. In the air data computer system
   b. In the autoflight or central warning system
   c. In the ATC transponder system

6. The indications given shortly before the desired height is reached are ●
   a. A white box and a steady chime
   b. An amber lamp and a short sound
   c. A red lamp and a continuous horn

7. The indications given in the approach mode are ●
   a. A repeated chime and a continuously illuminated lamp
   b. A short continuous sound and a flashing lamp
   c. A short continuous sound and a continuously illuminated lamp

8. An aircraft is climbing towards its preset altitude in the approach mode. When does the altitude alert end?
   - a. When the preset altitude is reached exactly
   - b. When the preset altitude is 900 ft away
   - c. When the preset altitude is 300 ft away

9. The altitude alert lamp is flashing and a short continuous sound is heard. Which mode is active?
   - a. The approach mode
   - b. The deviation mode
   - c. The capture mode

10. The landing gear is selected down. What is the state of the altitude alert system?
    - a. Inoperative
    - b. In the approach mode
    - c. In the deviation mode

11. The altitude is not in the alert range. Which mode is the digital altitude alert system in?
    - a. Approach
    - b. Capture
    - c. Reset

12. Where is the altitude alert computer located
    - a. In the master monitor card
    - b. In the air data computer
    - c. In the crew alerting card

13. The signals the altitude alert system receives from the AP/FD mode control panel are
    - a. Glideslope capture, parking brake status and selected ADC
    - b. Selected altitude, baro corrected altitude and FCC communication
    - c. Selected altitude, glideslope capture and FCC communication

14. The information the altitude alert system takes from the parking brake system is
    - a. That the parking brake is set
    - b. That the parking brake accumulator is charged
    - c. That the parking brake is not set

15. The digital altitude alert system enters the approach mode. What does the crew see on the PFD?
    - a. A flashing amber lamp beside the selected altitude
    - b. A white box around the current altitude
    - c. A white box around the selected altitude

16. The conditions for the reset mode of the digital altitude alert system are
    - a. Deviation greater than 300 ft and approach less than 900 ft
    - b. Deviation greater than 900 ft and approach greater than 900 ft
    - c. Deviation greater than 900 ft and approach less than 300 ft

17. A ground radar paints an aircraft. What can air traffic control establish from the radar alone?
    - a. The direction and the distance of an aircraft
    - b. The altitude and the distance of an aircraft
    - c. The direction and the altitude of an aircraft

18. Why is an ATC transponder installed in the aircraft ●

- **a.** Because for normal civil air surveillance the altitude cannot be found out directly
- **b.** Because the ground station cannot determine the direction of the aircraft
- **c.** Because radar cannot find out the distance of the aircraft directly

19. The sources from which the ATC transponder receives altitude information are ●

- **a.** An encoding altimeter or a radio altimeter
- **b.** An encoding altimeter or an air data computer
- **c.** An air data computer or the altitude alert computer

LEVEL 3 — APPLY AND REASON

20. After a deviation warning the aircraft goes on drifting away from the original altitude. At what point does the alert end? ●

- **a.** When the aircraft reaches the altitude selected on the control panel
- **b.** When the aircraft is more than 900 ft from the original altitude
- **c.** When the aircraft is more than 300 ft from the original altitude

21. Why is the altitude alert system deactivated by flap or landing gear extension on approach ●

- **a.** Because the altitude will then be constantly changing
- **b.** Because the ATC transponder takes over the altitude reporting
- **c.** Because the flap and gear signals interfere with the alert computer

22. The crew changes ALT SEL on the control panel during the cruise. What is the effect on the altitude alert system? ●

- **a.** The system gives an approach warning
- **b.** The system goes into deviation mode
- **c.** The system is inoperative

23. An aircraft is 1200 ft above its selected altitude. What indication does the altitude alert system give?

- **a.** Approach warning
- **b.** No warning
- **c.** Deviation warning

24. An aircraft is climbing towards its selected altitude and passes 700 ft below it. Which warning is given?

- **a.** No warning
- **b.** Deviation warning
- **c.** Approach warning

25. An aircraft established at the selected altitude drifts down until it is 500 ft below it. Which warning is given?

- **a.** No warning
- **b.** Approach warning
- **c.** Deviation warning

26. A technician checks when the altitude alert is inhibited. Which two conditions do that?

- **a.** The aircraft being in landing configuration, or the parking brake being set
- **b.** The aircraft being in landing configuration, or knob rotation being detected
- **c.** Knob rotation being detected, or the chime having already sounded

27. The altitude readout box on the PFD changes from white to yellow. Which mode has the system entered? ●

- **a.** Capture mode
- **b.** Deviation mode
- **c.** Approach mode

28. An altitude deviation occurs in the cruise. What does the crew get on EICAS and aurally? ●

- **a.** An advisory message ALTITUDE ALERT with no aural indication
- **b.** The warning message ALTITUDE ALERT with Master Warning and a bell
- **c.** The caution message ALTITUDE ALERT with Master Caution and the owl tone

29. Which configuration inputs does the crew alerting card use to inhibit the altitude alert?

- **a.** Landing gear down and locked, and the parking brake set
- **b.** Flaps in range, and speedbrakes stowed
- **c.** Flaps in range, and landing gear down and locked

30. A ground station transmits an interrogation to an aircraft. What does the aircraft equipment do? ●

- **a.** The ATC transponder answers with a reply that contains the altitude
- **b.** The ATC transponder answers with a reply that contains the distance
- **c.** The encoding altimeter answers with a reply that contains the altitude

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Answers — Altitude reporting and alerting

1. **c** The analogue system distinguishes an approach and a deviation warning. (L1, PDF p. 563)

2. **b** The approach mode is triggered at an altitude difference of 900 ft. (L1, PDF p. 563)

3. **c** Four modes are listed: approach, deviation, capture and reset. (L1, PDF p. 567)

4. **c** The deviation mode works on a deviation of 300 ft from the selected altitude. (L1, PDF p. 567)

5. **b** It normally sits inside the autoflight or central warning system, though a dedicated computer also exists. (L2, PDF p. 563)

6. **b** The approach information is an amber lamp with a short sound. (L2, PDF p. 563)

7. **c** In approach mode the lamp is steady, not flashing. (L2, PDF p. 563)

8. **c** The approach alert stops 300 ft short of the preset altitude. (L2, PDF p. 563)

9. **b** A flashing lamp distinguishes deviation from approach, where the lamp is steady. (L2, PDF p. 563)

10. **a** A gear down condition is one of the INOP conditions of the altitude alert system. (L2, PDF p. 563)

11. **b** Capture is the mode for the altitude not being in the alert range. (L2, PDF p. 567)

12. **c** The function lives on the crew alerting card. (L2, PDF p. 567)

13. **c** Those three signals are the mode control panel inputs. (L2, PDF p. 567)

14. **c** The input is the parking brake not set condition. (L2, PDF p. 567)

15. **c** Approach mode draws a white box around the selected altitude. (L2, PDF p. 567)

16. **c** Reset applies for a deviation beyond 900 ft and for an approach closer than 300 ft. (L2, PDF p. 567)

17. **a** Radar alone gives direction and distance, not altitude. (L2, PDF p. 571)

18. **a** The transponder exists because altitude is the quantity surveillance radar cannot get directly. (L2, PDF p. 571)

19. **b** Either an encoding altimeter or an air data computer supplies the transponder. (L2, PDF p. 571)

20. **b** The deviation alert ends either on regaining the altitude or on passing 900 ft away from it. (L3, PDF p. 563)

21. **a** A continuously changing altitude on approach would otherwise give constant alerts. (L3, PDF p. 563)

22. **c** Changing the selected altitude makes the system inoperative rather than producing a warning. (L3, PDF p. 563)

23. **b** Beyond 900 ft either side of the selected altitude the profile shows no warning. (L3, PDF p. 564)

24. **c** Closing on the selected altitude inside the 300 to 900 ft band gives the approach warning. (L3, PDF p. 564)

25. **c** Leaving the selected altitude into the 300 to 900 ft band gives the deviation warning. (L3, PDF p. 564)

26. **b** Either the landing configuration or a detected knob rotation inhibits the alert. (L3, PDF p. 565)

27. **b** A white to yellow change of the altitude readout box marks the deviation mode. (L3, PDF p. 567)

28. **c** It is a caution level alert with Master Caution and an owl tone. (L3, PDF p. 567)

29. **c** The crew alerting card takes flaps in range and landing gear down and locked; the parking brake comes in as a not set condition. (L3, PDF p. 570)

30. **a** The transponder replies to the interrogation, and the reply carries the altitude. (L3, PDF p. 571)

Clocks and time sources

PDF pages 575–584 · 12 questions

LEVEL 1 — RECALL

31. The primary function of the clock is ●

a. To display the elapsed time and the chronograph time
b. To give a time reference to other systems and display UTC
c. To supply the altitude alert system with a time reference

32. The secondary function of the clock is ●

a. To supply a time reference to the flight management computers
b. To give a display of date and year
c. To give a display of elapsed time or chronograph time

33. Which information does the on board maintenance system store so that repairs can be sped up ●

a. faults reported by the computers of various aircraft systems
b. parameters taken from the sensors of the flight controls
c. voices and sounds picked up in the cockpit during flight

34. In most aircraft that carry an on board maintenance system, in which ATA chapter is that system filed ●

a. ATA chapter 31
b. ATA chapter 23
c. ATA chapter 45

LEVEL 2 — UNDERSTAND AND EXPLAIN

35. An operator asks whether the voice recorder and the flight data recorder may be left out of an aircraft to save weight. What is the position? ●

a. Only the voice recorder is mandatory, the data recorder is optional
b. Both systems are mandatory and must be fitted
c. Both systems are optional and may be left out

36. Which ATA chapter holds both the flight data recording system and the monitoring system ●

a. chapter 23
b. chapter 31
c. chapter 45

37. What does the engineering department use a monitoring system to find out ●

a. which aircraft systems reported a fault during the last flight
b. which parts are in a critical condition and risk a malfunction
c. what the crew said to each other before an accident happened

38. A trainee reads HUMS in a helicopter wiring diagram and asks what kind of system it is. What is the answer? ●

a. It is one of the names used for the voice recorder system
b. It is one of the names used for the monitoring system
c. It is one of the names used for the maintenance system

39. What can the flight recorder system and the monitoring system have in common when both are fitted to one aircraft type ●

- **a.** They share some components
- **b.** They share one cockpit area microphone
- **c.** They share one crash protected memory

LEVEL 3 — APPLY AND REASON

40. Engineering wants an engine left on the wing as long as it is sound, yet changed at the home base before it causes trouble. Which installed system makes that possible? ●

- **a.** The on board maintenance system, which stores reported faults
- **b.** The monitoring system, which records data from various systems
- **c.** The flight data recording system, which records for the investigators

41. A part is drifting towards a critical condition while the aircraft is still away from base. How can engineering learn of it before the aircraft lands? ●

- **a.** The flight data recorder plays its data back to them over the voice radio
- **b.** The monitoring data can be sent to them in flight over a data transmission system
- **c.** The monitoring data is held in the recorder until the aircraft reaches the home base

42. Two of the four systems are fed from both sensors and system computers. Which pair is that?

- **a.** The flight data recording system and the monitoring system
- **b.** The cockpit voice recorder and the monitoring system
- **c.** The on board maintenance system and the voice recorder

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Answers — Clocks and time sources

31. **b** Time reference to other systems plus the UTC display is the stated primary function. (L1, PDF p. 575)

32. **c** Elapsed and chronograph time are the secondary function. (L1, PDF p. 575)

33. **a** The on board maintenance system stores the faults that the system computers report to it. (L1, PDF p. 583)

34. **c** Chapter 45 is the usual filing for the on board maintenance system, chapter 31 only in some aircraft. (L1, PDF p. 583)

35. **b** Both the cockpit voice recorder and the flight data recording system are mandatory, so neither may be left out. (L2, PDF p. 583)

36. **b** Both the flight data recording system and the monitoring system are filed in chapter 31. (L2, PDF p. 583)

37. **b** The monitoring system exists to show engineering which parts are close to failing. (L2, PDF p. 583)

38. **b** HUMS is listed as one of the manufacturers' names for the monitoring system. (L2, PDF p. 583)

39. **a** In some aircraft types the flight recorder system and the monitoring system share some components. (L2, PDF p. 583)

40. **b** Only the monitoring system records condition data that lets a part be used to its limit and still be changed at base. (L3, PDF p. 583)

41. **b** Monitoring data is not only stored, it can also be transmitted to engineering during the flight. (L3, PDF p. 583)

42. **a** Only the flight data recording system and the monitoring system take both sensors and system computers as sources. (L3, PDF p. 584)

Flight data recorders

PDF pages 585–595 · 38 questions

LEVEL 1 — RECALL

43. At which point does the flight data recorder start recording automatically ●

a. when an engine is started, or at the latest when the aircraft is in the air
b. when the aircraft is in the air, or at the latest when the gear is retracted
c. when the parking brake is released, or at the latest when an engine is started

44. The engines have just been switched off on the ground. How much longer does the flight data recording run? ●

a. 5 minutes later
b. 25 minutes later
c. 10 minutes later

45. What does the acquisition unit of a digital flight data recorder system receive ●

a. various data from aircraft systems and parameters from some sensors
b. the switching signals that come from the logic and the panel
c. the recorded data played back from the crash protected memory

46. With which standard does the digital flight data recorder comply ●

a. ARINC 573
b. ARINC 429
c. ARINC 717

47. In what does the digital flight data recorder store all aircraft information ●

a. removable PCMCIA memory cards
b. CMOS bulk erasable EEPROM devices
c. polyester magnetic tape cassettes

48. With which unit does the solid state flight recorder have a direct interface ●

a. the centralised data acquisition module
b. the flight warning computer
c. the flight management computer

49. In which form is the transmission data sent from the acquisition module to the recorder ●

a. pulse code modulation
b. frequency shift keying
c. pulse width modulation

LEVEL 2 — UNDERSTAND AND EXPLAIN

50. What does the transmitter fitted to the flight data recorder make possible ●

a. Marking the beginning of each radio call
b. Sending the recorded parameters to the ground
c. Locating it under water

51. What governs the installation of a flight data recorder in a large aeroplane

a. the relevant construction regulations such as CS-25 or FAR-25
b. the recording leaflet issued as JAA Leaflet No.19
c. the operating requirement Part - CAT CAT.GEN.MPA.195

52. With the approval of the responsible authority, greater acceptance regarding recording parameters is given to flight recorder systems installed

- **a.** after 01.04.2000
- **b.** before 01.04.98
- **c.** before 01.01.87

53. Which recorder is digital recording technology quickly putting in place of the conventional tape recorder ●

- **a.** the digital airborne recorder
- **b.** the solid state flight data recorder
- **c.** the quick access recorder

54. From where does the acquisition unit receive its date and leg information ●

- **a.** the flight management computer
- **b.** the data management unit
- **c.** the on board maintenance computer

55. Why does the acquisition unit carry a different name on some aircraft types?

- **a.** Because the unit is then filed in another ATA chapter
- **b.** Because the unit is then supplied by another manufacturer
- **c.** Because the unit might have additional functions there

56. Which parameters does the accelerometer provide to the recording system ●

- **a.** angular rate about all three axis
- **b.** linear acceleration along all three axis
- **c.** pitch and roll attitude about two axis

57. What follows from the digital flight data recorder being a solid state device ●

- **a.** It needs no built-in test equipment
- **b.** It has no moving parts
- **c.** It needs no electrical power

58. Which unit collects the data that the digital flight data recorder holds for the last 25 hours ●

- **a.** the CVR
- **b.** the FDIU
- **c.** the DMU

59. What maintenance does the underwater locator beacon call for in service? ●

- **a.** Replacement of its memory at set times
- **b.** Replacement of its battery at set times
- **c.** A functional transmission test at every flight

60. How many parameters in total can be saved by the recording system ●

- **a.** 1891
- **b.** 717
- **c.** 1024

61. How long can each word sent to the solid state flight data recorder be ●

- **a.** 8 bits long
- **b.** 16 bits long
- **c.** 12 bits long

LEVEL 3 — APPLY AND REASON

62. An aeroplane with a take-off weight under 5700 kg is being certified. How much recording capacity does its flight data recorder need? ●

- **a.** 10 hours
- **b.** 5 hours
- **c.** 25 hours

63. An engineer must decide how much data capacity a recorder needs for an aircraft of a given weight, seat count and certification date. Which rule governs that?

- **a.** The recording approval given in JAA Leaflet No.19
- **b.** Part - CAT CAT.GEN.MPA.195, or FAR 121.343
- **c.** The construction regulation CS-25, or FAR-25

64. An aeroplane weighs more than 27,000 kg and was certified before 01.01.87. Which further parameters must its recorder take?

- **a.** Each voice radio transmission, pitch attitude and roll attitude
- **b.** Thrust of each engine, air temperature and angle of attack
- **c.** Pitch trim, radio altitude and landing gear position

65. An aeroplane weighs more than 27,000 kg and was certified after 01.01.87. Which parameter group is then added on top of the rest?

- **a.** Pitch attitude, roll attitude and the start of each radio transmission
- **b.** Primary flight control position, pitch trim, radio altitude and gear position
- **c.** Thrust of each engine, air temperature and auto flight control mode

66. A recorder is set to begin taking data the moment the aircraft starts to taxi under its own power. Which requirement does that miss? ●

- **a.** It must record only once the aircraft has left the ground
- **b.** It must record from before the aircraft moves independently
- **c.** It must record from engine start until the brake is set

67. Why did the rising FAA parameter demands push manufacturers away from the tape recorder?

- **a.** Tape recorders could not be protected against thermal stresses
- **b.** Tape recorders were no longer accepted for aircraft above 5700 kg
- **c.** Recording that many parameters would demand too much from a tape recorder

68. A recorder has failed and cannot be repaired or replaced where the aircraft stands. How many subsequent flights may the aircraft make? ●

- **a.** No more than twelve subsequent flights
- **b.** No more than four subsequent flights
- **c.** No more than eight subsequent flights

69. A recorder failed 70 hours ago and the aircraft has flown three sectors since. What is the time limit that still has to be watched? ●

- **a.** No more than 90 hours may be exceeded following the failure
- **b.** No more than 72 hours may be exceeded following the failure
- **c.** No more than 25 hours may be exceeded following the failure

70. An aircraft is to be dispatched with a defective air data recorder. What condition is placed on the cockpit voice recorder? ●

- **a.** It must be operative, provided it is combined with the flight recorder
- **b.** It must be operative, provided it is not combined with the flight recorder
- **c.** It may be inoperative, provided the repair is done within eight flights

71. A line station is able to fix the recorder where the aircraft stands but wants to dispatch under the eight flight allowance. Is that allowed? ●

- **a.** Yes, that allowance holds whenever the recorder is defective
- **b.** No, that allowance needs repair on the spot to be impossible
- **c.** Yes, that allowance holds for 72 hours from the failure

72. A helicopter is on the ground with its rotor turning at 60%. What does the recording logic do?

- **a.** It provides the recorder with power, because rotor speed is above 50%
- **b.** It provides power only once the panel switch has been used
- **c.** It withholds power, because the rotor is turning on the ground

73. The aircraft is parked with its engines off and a technician needs the recorder running for a check. How can that be achieved?

- **a.** Through the panel, whose switch overrides the logic
- **b.** Through the accelerometer, which triggers the recorder on movement
- **c.** Through the acquisition unit, which commands the recorder directly

74. How is it confirmed that the recorder has actually written the data correctly?

- **a.** The recorder sends the recorded data back to the acquisition unit
- **b.** The panel compares the logic signal against the recorder status
- **c.** The acquisition unit compares its own data against the aircraft systems

75. An operator wants more than 25 hours of flight parameters held in the recorder. What makes that possible? ●

- **a.** Using the right combination of capacity and data rate
- **b.** Using a second recorder in parallel with the first
- **c.** Using a shorter word than twelve bits

76. The beacon battery is due for replacement. What has to be done to the recorder first? ●

- **a.** The recorder must be removed so the beacon can be taken off its panel
- **b.** Nothing, the beacon can be serviced without disassembly of the recorder
- **c.** The recorder memory must be downloaded before the beacon is opened

77. A frame is made up of four sub frames. How long does one frame take to record?

- **a.** 1 second
- **b.** 16 seconds
- **c.** 4 seconds

78. The identification data is spread out rather than held complete in one frame. How is it laid down?

- **a.** Distributed over several frames, usually in a specified word of a sub frame
- **b.** Written once at the start of recording, in the first frame only
- **c.** Repeated in every sub frame, always in the first word of the frame

79. Why is one parameter written several times within a frame while another appears only once?

- **a.** Because it is more important or changes more quickly
- **b.** Because it is carried in a discrete rather than a word
- **c.** Because it is a mandatory parameter required by law

80. Why do some aviation authorities require a quick access recorder at all, when the flight data recorder holds the same data? ●

- **a.** Because the copy of flight data then survives a crash better
- **b.** Because the copy of flight data then holds more parameters
- **c.** Because the copy of flight data then stays easily accessible

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Answers — Flight data recorders

43. **a** Engine start is the trigger, and being airborne is the latest point at which recording must have begun. (L1, PDF p. 585)

44. **a** The run-on time after engine shutdown is 5 minutes. (L1, PDF p. 585)

45. **a** Its inputs are the aircraft system data together with sensor parameters. (L1, PDF p. 589)

46. **c** The standard named for the DFDR is ARINC 717. (L1, PDF p. 591)

47. **b** The storage medium of the DFDR is CMOS bulk erasable EEPROM. (L1, PDF p. 591)

48. **a** Its direct interface is with the CDAM. (L1, PDF p. 595)

49. **a** The data reaches the recorder as pulse code modulation. (L1, PDF p. 595)

50. **c** The stated purpose of the transmitter is underwater location of the recorder. (L2, PDF p. 585)

51. **a** Installation falls under the construction regulations, CS-25 or FAR-25 for large aeroplanes. (L2, PDF p. 585)

52. **b** The relaxation covers recorder systems installed before 01.04.98. (L2, PDF p. 587)

53. **b** It is the SSFDR that is replacing the tape recorder. (L2, PDF p. 587)

54. **a** Date and flight leg come to the acquisition unit from the flight management computer. (L2, PDF p. 589)

55. **c** Extra functions on some types are the reason given for the different name. (L2, PDF p. 589)

56. **b** The accelerometer delivers linear acceleration in three axes. (L2, PDF p. 589)

57. **b** Solid state construction means there are no moving parts in the recorder. (L2, PDF p. 591)

58. **b** It is the FDIU whose collected data fills the 25 hours of recorder storage. (L2, PDF p. 591)

59. **b** The scheduled task on the beacon is battery replacement. (L2, PDF p. 591)

60. **a** A total of 1891 parameters can be saved. (L2, PDF p. 595)

61. **c** Word length under ARINC 717 is 12 bits. (L2, PDF p. 595)

62. **a** Below the 5700 kg take-off weight the capacity required drops to 10 hours. (L3, PDF p. 585)

63. **b** Capacity against weight, seats and certification date is set by the operating requirement, not the construction regulation. (L3, PDF p. 585)

64. **a** Those three are the extra parameters listed for the heavier aircraft certified before that date. (L3, PDF p. 585)

65. **b** That heavier post-87 group adds the flight control positions, pitch trim, radio altitude and gear position. (L3, PDF p. 585)

66. **b** The requirement reaches back to before the aircraft moves on its own, so recording cannot start at taxi. (L3, PDF p. 585)

67. **c** Recording the number of parameters the FAA came to demand would ask too much of a tape recorder, which is what accelerated the change to solid state. (L3, PDF p. 587)

68. **c** The dispatch relief is limited to eight subsequent flights. (L3, PDF p. 587)

69. **b** The elapsed time condition on dispatch is 72 hours after the failure. (L3, PDF p. 587)

70. **b** The voice recorder has to be serviceable, and that condition applies where it is a separate unit. (L3, PDF p. 587)

71. **b** The first condition is that repair or replacement cannot be done where the aircraft is. (L3, PDF p. 587)

72. **a** On helicopters the logic powers the recorder once rotor speed passes 50%. (L3, PDF p. 589)

73. **a** The panel switch overrides the logic so the recorder can be powered on the ground. (L3, PDF p. 589)

74. **a** Correct recording is monitored by returning the recorded data to the acquisition unit. (L3, PDF p. 589)

75. **a** Storage beyond 25 hours comes from matching recorder capacity with data rate. (L3, PDF p. 591)

76. **b** The beacon is serviceable in place, with no need to take the recorder apart. (L3, PDF p. 591)

77. **c** Four sub frames of one second each give a frame time of four seconds. (L3, PDF p. 593)

78. **a** The super frame spreads that data across several frames in a set word position. (L3, PDF p. 593)

79. **a** Repetition within a frame follows the importance of the parameter and how fast it changes. (L3, PDF p. 593)

80. **c** Easy access to the copy is the stated reason for the requirement. (L3, PDF p. 595)

Cockpit voice recorders

PDF pages 597–603 · 19 questions

LEVEL 1 — RECALL

81. What is the single unit called that combines the new generation voice and flight data recording of small aircraft ●

a. the cockpit voice flight data recorder
b. the combined flight data interface recorder
c. the cockpit voice data acquisition recorder

82. When does the cockpit voice flight data recorder start recording ●

a. when the rotor rpm exceeds 50%
b. when the control panel test switch is pressed
c. when electrical power is applied

83. A Class II data recorder is fitted to a civil commercial aircraft. Where is it mainly located? ●

a. the tail section
b. the forward cargo hold
c. the cockpit floor area

84. Where on a helicopter is the flight data recorder usually found ●

a. the nose compartment or the cockpit roof
b. one of the aft compartments or the tailboom
c. one of the engine bays or the main gearbox deck

85. At which frequency does the underwater locator beacon send out its audio signal ●

a. 40 MHz
b. 40 Hz
c. 40 kHz

LEVEL 2 — UNDERSTAND AND EXPLAIN

86. By which other name is the inertia sensor of the helicopter recording installation known

a. the area microphone
b. the immersion sensor
c. the accelerometer

87. When does the flight data recording within the combined unit start ●

a. When the area microphone picks up sound
b. When the electrical power is applied
c. When the rotor rpm exceeds 50%

88. What does the control panel switch of the combined voice and flight data recorder allow to be done ●

a. a memory download on ground
b. a battery change on ground
c. a BITE test on ground

89. Besides combining all data to a recordable format, what else does the flight data acquisition unit do ●

a. It monitors the recording
b. It erases the oldest data
c. It allocates the flight leg

90. Which condition stops both the voice recording and the flight data recording of the combined unit ●

- **a.** the voice erase button being pressed
- **b.** the rotor rpm falling below 50%
- **c.** the electrical power being disconnected

91. On aircraft of which weight is a single unit holding both the flight recorder and the cockpit voice recorder permitted ●

- **a.** above 27,000 kg
- **b.** below 5700 kg
- **c.** above 5700 kg

92. On what do the possible installation positions for a flight data recorder depend

- **a.** the classification of the recorder
- **b.** the number of parameters recorded
- **c.** the capacity of the recorder memory

93. What makes the helicopter recorder easy to pick out at a crash site? ●

- **a.** It is painted matt black with white stripes
- **b.** It is painted bright yellow or orange
- **c.** It is painted bright red with a flashing lamp

94. Which equipment picks up the signal from the locator beacon ●

- **a.** a shipboard radio receiver
- **b.** an underwater microphone
- **c.** a cockpit area microphone

95. For what minimum period does the beacon transmission continue once it has started ●

- **a.** 30 days
- **b.** 90 days
- **c.** 25 days

LEVEL 3 — APPLY AND REASON

96. The combined recorder has been running for eleven hours. What has become of its earliest hour of flight data? ●

- **a.** It has been kept, because recording stops once the memory is full
- **b.** It has been overwritten, because the oldest recording goes first
- **c.** It has been transferred into the voice section of the memory

97. The rotor has run down below 50% but the electrical power is still on. Which recording is still running? ●

- **a.** Neither recording, because the rotor has fallen below fifty per cent
- **b.** The flight data recording, while the voice recording has stopped
- **c.** The voice recording, while the flight data recording has stopped

98. Why is the helicopter recorder put in the aft compartments or the tailboom rather than further forward? ●

- **a.** Because those areas are easier to reach for a battery change
- **b.** Because those areas are further from the electrical interference
- **c.** Because those areas are less damaged in case of a crash

99. A recorder is lying on the sea bed after a ditching and nobody has touched it. What has set its beacon transmitting? ●

- **a.** The loss of aircraft electrical supply to the recorder
- **b.** Its battery coming in contact with water
- **c.** The impact sensed by the inertia sensor of the recorder

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Answers — Cockpit voice recorders

81. **a** The combined unit is the CVFDR. (L1, PDF p. 597)

82. **c** Application of electrical power is what starts the unit recording. (L1, PDF p. 597)

83. **a** The usual position for a Class II recorder is the tail section. (L1, PDF p. 599)

84. **b** Those are the two areas named for the helicopter installation. (L1, PDF p. 601)

85. **c** The beacon signal is at 40 kHz. (L1, PDF p. 603)

86. **c** The inertia sensor of the helicopter recording installation is the accelerometer. (L2, PDF p. 597)

87. **c** Flight data recording is tied to rotor speed above 50%. (L2, PDF p. 597)

88. **c** The panel test switch runs a BITE test on the ground. (L2, PDF p. 597)

89. **a** Besides formatting the data the unit monitors the recording. (L2, PDF p. 597)

90. **c** Disconnecting the electrical power stops the unit recording; falling below 50% rotor rpm stops only the flight data part. (L2, PDF p. 597)

91. **b** The combined unit is allowed below the 5700 kg mark. (L2, PDF p. 599)

92. **a** Where the recorder may go follows from its classification. (L2, PDF p. 599)

93. **b** Bright yellow or orange paint is what makes it easy to find. (L2, PDF p. 601)

94. **b** An underwater microphone is what receives the beacon. (L2, PDF p. 603)

95. **b** The minimum transmission period is 90 days. (L2, PDF p. 603)

96. **b** Once the holding time is full the oldest data is overwritten. (L3, PDF p. 597)

97. **c** Flight data stops with rotor speed below 50%, the voice recording runs until power is removed. (L3, PDF p. 597)

98. **c** Crash damage is lower there, which is why the recorder is placed aft. (L3, PDF p. 601)

99. **b** Water reaching the battery starts the transmission automatically. (L3, PDF p. 603)

Condition monitoring, quick access recorders and HUMS

PDF pages 605–615 · 37 questions

LEVEL 1 — RECALL

100. Which three areas does the aircraft integrated data system monitor and analyse continuously ●

a. engine data, aircraft data and gearbox data
b. engine data, aircraft data and APU data
c. engine data, voice data and flight data

101. What does APM stand for in the aircraft condition monitoring system ●

a. aircraft parameter monitoring
b. auxiliary power monitoring
c. aircraft performance monitoring

102. Which data does the quick access recorder record ●

a. the same data as the digital flight data recorder
b. only the data that the data management unit programmes
c. a shortened selection of the flight data recorder data

103. About how long does a quick access recorder write before its recording time is used up? ●

a. 10 hours
b. 25 hours
c. 50 hours

104. Besides recording and storing digital flight data, what is the wireless quick access recorder there to do ●

a. transmit digital flight data without wires
b. erase digital flight data after transfer
c. format digital flight data for the recorder

105. A technician has to take the recorded data off a wireless quick access recorder. On which medium is it held? ●

a. a polyester magnetic tape cassette
b. a standard PCMCIA removable PC Card
c. a bulk erasable EEPROM device

106. In the flight avionics of which aircraft are health and usage monitoring systems a common component ●

a. a modern large aeroplane
b. a modern business jet
c. a modern helicopter

107. Exact data about the manoeuvres flown has been recorded. Which additional usage parameter is that? ●

a. gross weight
b. flight envelope
c. flight endurance

LEVEL 2 — UNDERSTAND AND EXPLAIN

108. Which unit manages the monitoring and analysis tasks in the ACMS ●

a. the centralised data acquisition module
b. the flight data interface unit
c. the data management unit

109. Which further task does the data management unit take on with its smart airborne recorder or wireless quick access recorder ●

a. checking the mandatory parameters for integrity
b. recording pre-programmed parameters
c. erasing the oldest recorded parameters

110. What else can be done with the data of all the systems connected to the data management unit? ●

a. It can be retrieved for trouble-shooting
b. It can be erased before the next flight
c. It can be written to the crash protected memory

111. Two aircraft carry the same monitoring system but one calls it ADAS and the other ACMS. How does that come about? ●

a. The system name is given by the manufacturer
b. The system name is given by the ATA chapter it is filed in
c. The system name is given by the certifying authority

112. From where does the digital airborne recorder get the free programmed data it writes ●

a. the data management unit
b. the flight warning computer
c. the flight data interface unit

113. Where does the digital flight data handled by the wireless quick access recorder generally come from ●

a. the flight data interface and management unit
b. the flight warning computer
c. the centralised data acquisition module

114. When is the recorded data of the wireless quick access recorder compressed, encrypted and transmitted ●

a. while the aircraft is on the ground
b. while the aircraft is in the cruise
c. while the aircraft is being powered down

115. Over which link is the recorded flight information sent securely from the aircraft on the ground ●

a. a commercial wireless cellular data link
b. a satellite voice communication link
c. a wired ground support equipment link

116. What makes up the user interface of the wireless quick access recorder

a. a control display unit and a station printer
b. a front panel keypad and an alphanumeric display
c. a front panel test connector and a fail indicator lamp

117. On what do the parameters monitored by a health and usage monitoring system depend ●

a. the certification date of the helicopter and its weight
b. the flight time flown since the last scheduled inspection
c. the helicopter configuration, such as engines, gearboxes and rotors

118. What do speed probes enable maintenance to identify on rotating components ●

a. A component overpressure
b. A component overheat
c. A component overspeed

119. What can a monitored exceedance of temperature limits or a tendency to run hot be ●

a. the sign of a loss of vital system fluid
b. the beginning of a major component or system failure
c. the result of an overspeed of a rotating component

120. Why is vibration measured on bearings in particular? ●

a. Because they carry no temperature or pressure sensor of their own
b. Because they turn faster than the components around them
c. Because they are not inspected at preflight or are out of reach to inspect

121. What does a rising vibration reading on a monitored component indicate about that component ●

a. It is beginning to wear out and may cause severe damage
b. It is beginning to overspeed and may cause an overstress
c. It is beginning to run hot and may cause a fluid loss

122. How does the data used by usage monitoring compare with that used by health monitoring ●

a. the same set with some extensions
b. a separate set taken from other sensors
c. a reduced set limited to the rotor system

123. Which part of the operation do the parameters recorded for usage monitoring serve ●

a. the logistic and maintenance part
b. the flight planning and dispatch part
c. the accident investigation part

124. Flight endurance recorded by usage monitoring gives an exact helicopter run time instead of a time ●

a. rounded down to the last full hour
b. rounded up to the next 5 minutes
c. rounded up to the next 15 minutes

LEVEL 3 — APPLY AND REASON

125. Data has to reach the evaluation computer on the ground while the aircraft is still flying. Which interface carries it? ●

a. ADL
b. MPRT
c. ACARS

126. The ground computer is to be supplied with print reports through the station terminals. Which interface produces them? ●

a. ADL
b. ACARS
c. MPRT

127. The monitoring system is to be modified by loading from disks. Which of its interfaces is used? ●

a. ACARS
b. MCDU
c. ADL

128. A quick access recorder is coming to the end of its writing time. What must be done? ●

a. The recording medium must be erased before the 50 hours have elapsed
b. The recording medium must be changed before the 50 hours have elapsed
c. The recording medium must be downloaded before the 25 hours have elapsed

129. The digital airborne recorder runs at a faster recording speed than the quick access recorder. What follows from that? ●

a. Its maximum writing time is longer
b. Its maximum writing time is shorter
c. Its stored data is harder to access

130. A unit of this type is fitted with no radios at all. What is it then identified as?

a. A digital airborne quick access recorder
b. A PC Card quick access recorder
c. A wireless quick access recorder

131. A main gearbox has no cockpit instrument or warning of its own. Why does that make its speed probe reading so valuable? ●

a. Because a vibration there could otherwise be felt through the controls
b. Because an overheat there could otherwise be read on the oil gauge
c. Because an overspeed there could otherwise go unidentified by the crew

132. A monitored circuit shows a steady tendency towards low pressure. What does the monitoring read into that? ●

a. An upcoming failure or a loss of vital system fluid
b. An overstress of a component or abnormal wear
c. A bearing beginning to wear out in the gearbox

133. Acceleration readings come back higher than normal after a sortie. What does the monitoring system take that to mean? ●

a. A component has run hot or a system failure is beginning
b. A component has been overstressed or abnormal wear is occurring
c. A component has overspeeded or a bearing has failed

134. Metal particles are detected in a gearbox oil system. What does the monitoring system conclude from that? ●

a. A gear is overspeeding and could shed teeth if not slowed
b. A bearing is wearing abnormally and could fail if not checked
c. A bearing is running hot and could seize if not cooled

135. A component is limited by the safe life principle and its limit is approaching. How can usage monitoring help? ●

a. The time can be extended when no fatigue or damage has been recorded
b. The time can be extended when no overspeed has been reported by the crew
c. The limit can be removed once the component has been vibration checked

136. Gross weight is recorded and set against the recorded flight envelope. What is that comparison looking for? ●

a. Possible overstress to components and structural limits
b. Possible loss of vital fluid from a pressurised system
c. Possible excess of the safe life limit of a component

Answers — Condition monitoring, quick access recorders and HUMS

100. b The three listed are engine, aircraft and APU data. (L1, PDF p. 605)

101. c APM is aircraft performance monitoring. (L1, PDF p. 605)

102. a Its records are identical to those of the DFDR. (L1, PDF p. 607)

103. c The writing time of the quick access recorder is about 50 hours. (L1, PDF p. 607)

104. a Recording, storing and wireless transmission are the three stated purposes. (L1, PDF p. 613)

105. b It records on a standard PCMCIA removable PC Card. (L1, PDF p. 613)

106. c Health and usage monitoring systems are a common component in the flight avionics of modern helicopters. (L1, PDF p. 615)

107. b Flight envelope is the parameter carrying the manoeuvres flown. (L1, PDF p. 615)

108. c The DMU manages those tasks. (L2, PDF p. 605)

109. b Through those recorders the DMU records parameters that have been pre-programmed. (L2, PDF p. 605)

110. a That connected data is available for trouble-shooting. (L2, PDF p. 605)

111. a It is the manufacturer who names the system. (L2, PDF p. 605)

112. a Its free programmed data is supplied by the DMU. (L2, PDF p. 607)

113. a The FDIMU is the normal source of its data. (L2, PDF p. 613)

114. a The transfer takes place with the aircraft on the ground. (L2, PDF p. 613)

115. a A commercial cellular data link carries the transfer. (L2, PDF p. 613)

116. b The user works through a keypad and an alphanumeric display on the front panel. (L2, PDF p. 613)

117. c The configuration of the helicopter decides what gets monitored. (L2, PDF p. 615)

118. c The speed probes exist to catch an overspeed of a rotating component. (L2, PDF p. 615)

119. b Rising temperature is treated as the start of a major component or system failure. (L2, PDF p. 615)

120. c Bearings get the attention because a preflight inspection cannot see or reach them. (L2, PDF p. 615)

121. a Vibration growth indicates wear that could go on to damage the system badly. (L2, PDF p. 615)

122. a Usage monitoring works from the health monitoring data, extended a little. (L2, PDF p. 615)

123. a Those parameters are there for logistics and maintenance. (L2, PDF p. 615)

124. b The gain is an exact run time instead of rounding up to the next 5 minutes. (L2, PDF p. 615)

125. c The ground computer is fed through ACARS. (L3, PDF p. 605)

126. c The multi-printer interface is what delivers the print reports. (L3, PDF p. 605)

127. c Modification by disk goes through the airborne data loader. (L3, PDF p. 605)

128. b There is no erase button, so the recording medium has to be replaced inside the 50 hours. (L3, PDF p. 607)

129. b A faster recording speed fills the same medium sooner, so the writing time is shorter. (L3, PDF p. 607)

130. b Without radios the unit is the PQAR, the PC Card quick access recorder. (L3, PDF p. 613)

131. c Components the crew cannot watch are exactly the ones whose overspeed the probe would otherwise miss. (L3, PDF p. 615)

132. a A drifting pressure points to a coming failure or to fluid being lost. (L3, PDF p. 615)

133. b High acceleration is read as overstress or abnormal wear. (L3, PDF p. 615)

134. b Metal in the oil points to abnormal bearing wear that may end in failure. (L3, PDF p. 615)

135. a Recorded absence of fatigue or damage is what allows the safe life time to be extended. (L3, PDF p. 615)

136. a The comparison is made to find overstress of components and structural limits. (L3, PDF p. 615)

Gyroscopic principles: rigidity and precession

PDF pages 619–629 · 40 questions

LEVEL 1 — RECALL

137. On which three bases do the gyroscopes used in aviation today work ●

a. an optical basis, a thermal basis and an oscillation basis
b. a mechanical basis, an optical basis and an oscillation basis
c. a mechanical basis, a pneumatic basis and a magnetic basis

138. For which three flight instruments are the properties of a gyroscope used ●

a. the directional gyro, the altimeter and the turn-and-bank indicator
b. the artificial horizon, the airspeed indicator and the vertical speed indicator
c. the artificial horizon, the directional gyro and the turn-and-bank indicator

139. Around how many axes can any body move in free space ●

a. three
b. two
c. four

140. A gyro has one degree of freedom. What movement is open to it? ●

a. movement around any axis in free space
b. movement around an axis perpendicular to its own
c. movement around its own axis only

141. What do the letters ERP stand for ●

a. Earth Rotation Precession
b. Equator Rate Precession
c. Earth Rate Precession

142. What is the horizontal component of apparent wander called ●

a. apparent drift
b. apparent topple
c. transport wander

LEVEL 2 — UNDERSTAND AND EXPLAIN

143. What does a running gyroscope tend to do ●

a. maintain the direction of its axis in space
b. align its axis with the earth's magnetic field
c. hold its axis square to the aircraft structure

144. The inertia of a gyroscope against the forces acting on it increases with ●

a. rotational speed, bearing friction and frame weight
b. rotational speed, mass and distance from the rotational axis
c. mass, ambient temperature and distance from the rotational axis

145. Which two actions is a gyroscope free to accept without opposing them ●

a. a force at right angles to the axis and a parallel shift of the gyro axis
b. a force along the direction of the axis and a tilting of the gyro axis
c. a force along the direction of the axis and a parallel shift of the gyro axis

146. At what angle to an exterior force trying to change the direction of its axis does a gyroscope precess ●

a. 90°
b. 180°
c. 45°

147. On what does the direction in which a gyroscope precesses depend ●

a. the size of the exterior force
b. the rotation of the gyroscope
c. the mass of the gyroscope

148. A gyro running at medium speed is pulled out of its mounting anyway. What is the damage to be expected?

a. the bearings and the frame can be destroyed
b. the electrical drive winding can be burnt out
c. the gimbal springs can be stretched beyond repair

149. What is said of a body whose movement around one axis is limited ●

a. it has a low degree of freedom
b. it has a half degree of freedom
c. it has a total degree of freedom

150. What does a body that can move around all of its axes have ●

a. a high (total) degree of freedom
b. a low degree of freedom
c. a half degree of freedom

151. How is a gyroscope with two degrees of freedom mounted ●

a. in a frame which can itself rotate around an axis parallel to the gyro axis
b. in two frames which can rotate around axes perpendicular to each other
c. in a frame which can itself rotate around an axis perpendicular to the gyro axis

152. A gyroscope with three degrees of freedom is gimballed in cardan frames. How many cardan frames does it have? ●

a. three
b. one
c. two

153. A gyroscope can be set to any position at all in free space. What is such a gyroscope called? ●

a. a spring restrained gyroscope
b. a fully gimballed gyroscope
c. a rate gyroscope

154. By what means is a gyroscope frame constrained so that it finds a position of rest ●

a. by liquid level switches
b. by torque motors
c. by springs

155. Which degree of freedom does American literature leave out

a. the one around the spin axis
b. the one around the outer cardan frame
c. the one around the inner cardan frame

156. At what angle to each other are the frames of a fully gimballed gyro system set ●

a. 45°
b. 90°
c. 120°

157. Which two instruments are based on the space stability of the gyroscope ●

a. the artificial horizon and the vertical speed indicator
b. the directional gyro and the turn-and-bank indicator
c. the directional gyro and the artificial horizon

158. What causes the apparent wander of the axis of a free gyroscope that is called ERP ●

a. the east-west movement of the aircraft carrying the gyroscope
b. the friction of the bearings acting on the frames of the gyroscope
c. the rotation of the earth acting on the space stability of the gyroscope

159. Into which two components is the three dimensional wander divided in relation to the earth's surface ●

a. a horizontal and a vertical component
b. a longitudinal and a lateral component
c. a real and an apparent component

160. To what does apparent drift amount per hour ●

a. 15 deg cosine of latitude
b. 15 deg sine of latitude
c. 15 deg tangent of latitude

161. To what does apparent topple amount per hour ●

a. 15 deg cosine of latitude
b. 15 deg sine of latitude
c. 15 deg tangent of latitude

162. Which aircraft movement causes the apparent wander called transport wander ●

a. east-west movement
b. vertical movement
c. north-south movement

163. A fully gimballed vertical gyro stands at the equator and an observer watches it for a full day. Through how much does it appear to topple? ●

a. 360° in 24 hours
b. 180° in 24 hours
c. 360° in 12 hours

164. What happens to the apparent topple of a vertical gyro at higher latitudes ●

a. it becomes larger and is greatest at the poles
b. it becomes smaller and is zero at the poles
c. it stays the same at every latitude

165. Which way does the axis of a gyro point when apparent topple is being considered

a. along the aircraft's direction of flight
b. vertical related to the earth's surface
c. parallel to the earth's surface

LEVEL 3 — APPLY AND REASON

166. An exterior force has been applied to the axis of a running gyro and is held there. When does the gyro stop precessing? ●

a. when its sense of rotation corresponds to the rotational direction acting upon it
b. when its axis has turned through a right angle from the direction it started in
c. when the inertia of its rotating mass has fully absorbed the applied force

167. A gyroscope was switched off approximately three minutes ago and a technician wants to take it out now. What is the position?

a. it may be removed, because only a stationary rotor can be damaged
b. it may in no case be removed, because the gyro will tend to flounder
c. it may be removed, because the rotor has already come to rest

168. A gyroscope frame is constrained by means of springs. What is the freedom of that whole gyro system? ●

a. 2 1/2 degrees of freedom
b. 1 degree of freedom
c. 1 1/2 degrees of freedom

169. Why does a spring constrained frame count as only half a degree? ●

a. because it can move about only half of the axes that a free frame can
b. because it has a position of rest to which it constantly tries to return
c. because it is driven through only half of its travel before it is stopped

170. An aircraft rolls and pitches into a completely new attitude. What has happened to the axis of its space stable gyro? ●

a. it still points to the same direction
b. it has precessed through a right angle to the new attitude
c. it has followed the aircraft into the new attitude

171. The tail of a propeller aircraft is lifted during the take-off run and the pilot does not react. What does the nose do?

a. it turns to the right
b. it turns to the left
c. it rises further

172. The landing gear is being retracted while the wheels are still spinning. Why does that matter?

a. the spinning wheels drag the gyro instruments out of their rest position
b. the spinning wheels lose their space stability once they leave the airflow
c. the spinning wheels carry precession forces like any other rotating component

173. A free gyroscope is left standing at the equator. Which of the two apparent errors is at its greatest there? ●

a. apparent topple
b. apparent drift
c. transport wander

174. A free gyro is no longer standing at one place but is being carried along. Why do the plain earth rate figures no longer cover all of its apparent error? ●

a. because those figures hold only on the ground, where the bearings are unloaded
b. because those figures hold only with no aircraft movement, so transport wander is added
c. because those figures hold only for a gyro with a half degree of freedom

175. A gyro axis is seen to move although nothing mechanical is wrong with it. Under which heading does that movement fall?

- **a.** gimbal error, the indication error that builds up during a turn
- **b.** real wander, the precession motion set up by friction in the bearings
- **c.** apparent wander, the apparent motion relative to the rotating earth

176. Why does a gyro that holds its direction appear to topple to someone standing beside it at the equator?

- **a.** because the bearings of the gyro drag its axis round with the earth
- **b.** because gravity pulls the heavy end of the gyro axis downwards
- **c.** because the earth is spinning below the space stable gyroscope

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Answers — Gyroscopic principles: rigidity and precession

137. b The three families named are mechanical, optical and oscillation gyroscopes. (L1, PDF p. 619)

138. c Those three instruments are the ones named as built on gyroscopic properties. (L1, PDF p. 619)

139. a Any body can move around three axes in free space. (L1, PDF p. 621)

140. c With one degree of freedom the gyroscope can only move around its own axis. (L1, PDF p. 621)

141. c ERP is Earth Rate Precession. (L1, PDF p. 627)

142. a The horizontal component of apparent wander is apparent drift. (L1, PDF p. 627)

143. a The gyroscope makes every effort to maintain the direction of its axis in space. (L2, PDF p. 619)

144. b Inertia in relation to forces acting on the gyro increases with rotational speed, mass and distance from the rotational axis. (L2, PDF p. 619)

145. c The gyroscope resists neither a force acting along its axis nor a parallel shift of that axis. (L2, PDF p. 619)

146. a The gyroscope precesses at a 90 degree angle to the force acting on it. (L2, PDF p. 619)

147. b The direction of precession depends on the rotation of the gyroscope itself. (L2, PDF p. 619)

148. a A gyro removed at medium speed will flounder and the bearings and frame can be destroyed. (L2, PDF p. 619)

149. a Limiting the movement around one axis gives a low degree of freedom. (L2, PDF p. 621)

150. a Movement around all axes gives a high, that is total, degree of freedom. (L2, PDF p. 621)

151. c Two degrees of freedom come from one frame that can rotate about an axis perpendicular to the gyro axis. (L2, PDF p. 621)

152. c Three degrees of freedom are obtained from two frames, the inner and the outer cardan frame. (L2, PDF p. 621)

153. b A gyroscope which can assume any position in free space is a fully gimballed gyroscope. (L2, PDF p. 621)

154. c The frame is constrained by means of springs so that it has a rest position to return to. (L2, PDF p. 621)

155. a American literature does not count the degree of freedom around the spin axis. (L2, PDF p. 621)

156. b The three frames of the fully gimballed mounting stand at 90° to each other. (L2, PDF p. 623)

157. c Directional gyro and artificial horizon are the two space stability devices named. (L2, PDF p. 623)

158. c ERP is apparent wander caused by the rotation of the earth acting on the gyro's space stability. (L2, PDF p. 627)

159. a Relative to the earth's surface the wander splits into horizontal and vertical components. (L2, PDF p. 627)

160. b Apparent drift amounts per hour to 15° Sinus of latitude. (L2, PDF p. 627)

161. a Apparent topple amounts per hour to 15° cosinus of latitude. (L2, PDF p. 627)

162. a Transport wander is caused by the east-west movement of the aircraft. (L2, PDF p. 627)

163. a At the equator the gyroscope appears to topple 360° in 24 hours. (L2, PDF p. 629)

164. b Apparent topple becomes smaller at higher latitudes and is zero at the poles. (L2, PDF p. 629)

165. b Apparent topple is described for a fully gimballed vertical gyro whose axis is vertical to the earth's surface. (L2, PDF p. 629)

166. a The gyro precesses until its sense of rotation corresponds to the rotational direction acting upon it. (L3, PDF p. 619)

167. b Approximately three minutes after switching off the gyro is running at medium speed, and at medium speed it may in no case be removed. (L3, PDF p. 619)

168. c A spring constrained frame counts as a half degree, so the gyro system has 1 1/2 degrees of freedom. (L3, PDF p. 621)

169. b The spring gives the frame a position of rest, so it no longer has its full freedom of movement. (L3, PDF p. 621)

170. a When the aircraft changes attitude the gyro axis still points in the same direction. (L3, PDF p. 623)

171. b Lifting the tail applies a force to the propeller axis which is transferred clockwise into the plane of precession, so the nose turns left. (L3, PDF p. 625)

172. c Wheels turning during gear retraction are listed among the rotating components that carry precession forces. (L3, PDF p. 625)

173. a Apparent topple is greatest at the equator. (L3, PDF p. 627)

174. b The earth rate statements are only true if the gyroscope remains at the same place; once the aircraft moves, transport wander comes on top. (L3, PDF p. 627)

175. c Apparent wander is apparent movement caused by maintaining a position in space, relative to the rotating earth or by transport. (L3, PDF p. 628)

176. c The gyro keeps its direction in space while the earth spins below it, so the observer sees it topple. (L3, PDF p. 629)

Drift, topple and erection

PDF pages 631–641 · 37 questions

LEVEL 1 — RECALL

177. What kind of influences cause real drift and real topple ●

a. atmospheric influences
b. magnetic influences
c. constructional influences

178. By which other name is the gimbal error of the directional gyro known ●

a. the cardan error
b. the gimbal lock
c. the transport error

179. An erection system needs something to judge the earth's surface by. What does it use as a reference signal? ●

a. the magnetic field of the earth
b. the static pressure of the atmosphere
c. the gravity of the earth

180. Which two types of optical gyroscope are there ●

a. the ring laser gyro and the fibre optic gyro
b. the fibre optic gyro and the quartz oscillating gyroscope
c. the ring laser gyro and the hemispherical resonator gyro

181. Which parts does the ring laser gyroscope have ●

a. a double laser exciter, prisms, a fibre coil and a photodetector
b. a double laser exciter, mirrors, prisms and light sensors
c. a single laser exciter, mirrors, lenses and light sensors

182. What is interference ●

a. the overlapping of oscillations and waves
b. the cancelling of oscillations by waves
c. the splitting of oscillations into waves

LEVEL 2 — UNDERSTAND AND EXPLAIN

183. A fully gimballed directional gyro is over the North Pole and an observer turning with the earth watches it for a day. Through how much does it appear to rotate? ●

a. 360° in 24 hours
b. 360° in 12 hours
c. 180° in 24 hours

184. What happens to the apparent drift of a directional gyro at lower latitudes ●

a. it stays the same at every latitude
b. it becomes larger and is greatest at the equator
c. it becomes smaller and is zero at the equator

185. How does the axis of the directional gyro lie when apparent drift is being considered

a. at 45° to the earth's surface
b. vertical to the earth's surface
c. parallel to the earth's surface

186. Which two faults cause real drift and real topple ●

a. defective gyro and frame bearings, and the rotation of the earth
b. unbalance of the gyro and frame, and the east-west transport of the aircraft
c. defective gyro and frame bearings, and unbalance of the gyro and frame

187. A directional gyro is suffering from its gimbal error. During which manoeuvre does that mainly happen? ●

a. a turn
b. a sideslip
c. a climb

188. What does the gimbal effect in the directional gyro result in ●

a. a loss of rotor speed
b. a direction indication error
c. a jammed outer frame

189. What is the state of the two frames when gimbal lock has occurred ●

a. they are aligned
b. they are held apart by their springs
c. they are at 90° to each other

190. What must be eliminated before a gyro can be used for aircraft instruments ●

a. the total effect of topple
b. the whole of the spin axis friction
c. the half degree of freedom of the frame

191. What does the erection system control the gyro axis to keep ●

a. a constant direction to the earth's surface
b. a constant angle to the aircraft centre line
c. a constant speed of rotation about its own axis

192. By which two means is the erection reference signal detected ●

a. by a moving mass like a pendulum or by liquid level switches
b. by liquid level switches or by a pressure capsule
c. by a moving mass like a pendulum or by a magnetic flux valve

193. By which two means is the erection force applied to the gyro ●

a. by electrical torque motors or by air jets on the rotor
b. by mechanic force of gravity or by electrical torque motors
c. by mechanic force of gravity or by restraining springs

194. What occurs between two coherent beams of light running opposite ways on the same course if the apparatus is rotated

a. a phase displacement
b. a change of colour
c. a loss of coherence

195. An optical gyroscope is being explained to a trainee. According to which optical principle does it work? ●

a. the Doppler effect
b. the swirl effect
c. the Sagnac effect

196. On which principle does the ring laser gyroscope work ●

a. a single laser beam in a ring cavity is compared with a fixed reference
b. two counterpropagating laser beams in a ring cavity are added to each other
c. two counterpropagating laser beams in a ring cavity are compared with each other

197. Under which condition is light coherent ●

a. when it has been gathered by reflection from several light packets
b. when it has been filtered to one single wave length
c. when it has been split by reflection from one single light packet

LEVEL 3 — APPLY AND REASON

198. Two gyros stand at the North Pole, one with its axis upright and one with its axis lying along the ground. Which apparent error is at its maximum there? ●

a. drift, for the gyro whose axis is parallel to the surface
b. neither, both of those errors fall to zero at the pole
c. topple, for the gyro whose axis stands vertical to it

199. An artificial horizon gyro whose axis points to the centre of the earth at the north pole is carried down to the equator. How does its axis finish up to an observer on the earth?

a. horizontal
b. turned through a right angle in the horizontal plane
c. still vertical

200. A gyro has worn frame bearings and is also carried east-west across the earth. How do the two error sources add up? ●

a. the bearing error is cancelled by the earth rotation and transport errors
b. the bearing error takes the place of the earth rotation and transport errors
c. the bearing error sits on top of the earth rotation and transport errors

201. A gyroscope is found with its two frames lined up with each other. What can still be done with it? ●

a. it can be used once its frames are re-sprung
b. it can be used for drift but not for topple
c. nothing, an instrument in that state is unusable

202. An erection system is fitted to a directional gyro. In which direction does it hold that gyro's axis? ●

a. horizontal
b. vertical
c. along the aircraft centre line

203. What decides whether a given gyro is erected mechanically or by torque motors? ●

a. the latitude the aircraft is operating at
b. the type of gyro
c. the number of frames the gyro is gimballed in

204. An erection system is set to work against the total topple of a gyro. Which two errors make up that total?

a. real topple and real drift
b. apparent topple and real topple
c. apparent topple and apparent drift

205. A designer is choosing how big to make the beam cavity of an optical gyro. Why is a larger one wanted?

a. the laser needs less power to keep the beams going
b. the interference pattern can be measured more easily
c. the two beams stay in phase over a wider speed range

206. At extremely small angular velocities the beams of light still interfere. What is the difficulty then?

a. the interference is difficult to measure
b. the interference pattern no longer forms at all
c. the two beams stop running in opposite directions

207. An optical gyro is built with mirrors rather than with fibre optics. What does that choice bring with it?

a. easy measurement and a very large orbit
b. difficult measurement but a size constrained unit
c. easy measurement but a size constrained unit

208. The gyroscope block is turned clockwise. What happens to the beam that runs in the direction of that rotation?

a. it stops returning to its starting point at all
b. it travels slightly further before returning to its starting point
c. it travels slightly less far before returning to its starting point

209. One of the two beams has been forced onto the longer course by the rotation. What has become of its frequency?

a. it is unchanged
b. it is slightly lower
c. it is slightly higher

210. The two wave trains are keeping exactly the same frequencies. What are the bright and dark strips doing?

a. they wander over the photodiodes
b. they double in number
c. they remain stationary

211. The frequencies of the two wave trains begin to differ. What does the counter then receive?

a. alternating light-dark impulses as the strips wander over the photodiodes
b. a steady light level as the strips settle over the photodiodes
c. a single bright pulse as the strips cancel at the photodiodes

212. The photodiodes have told the electronics which way the interference strips are wandering. What is that used for?

a. the rotational direction, carried as a preceding symbol in the pulse count
b. the rotational speed, carried as a count of the pulses per unit of time
c. the path length, carried as a correction to the setting of the mirrors

213. Coherent light has been made to interfere inside the optical circuit of a gyro. What can then be measured? ●

a. the direction of the earth's gravity vector
b. the speed of rotation, that is the arc velocity
c. the temperature of the resonance cavity

Answers — Drift, topple and erection

177. c Real drift and real topple come from constructional influences. (L1, PDF p. 633)

178. a Gimbal error of the directional gyro is known as the cardan error. (L1, PDF p. 633)

179. c The erection system uses the gravity of the earth as its reference signal. (L1, PDF p. 635)

180. a The two optical types are the ring laser gyro and the fibre optic gyro. (L1, PDF p. 637)

181. b The RLG has a double laser exciter, mirrors, prisms and light sensors. (L1, PDF p. 637)

182. a Interference is the overlapping of oscillations and waves. (L1, PDF p. 641)

183. a At the pole the gyroscope appears to rotate 360° in 24 hours. (L2, PDF p. 631)

184. c Apparent drift becomes smaller at lower latitudes and is zero at the equator. (L2, PDF p. 631)

185. c Apparent drift is described for a fully gimballed directional gyro whose axis lies parallel to the earth's surface. (L2, PDF p. 631)

186. c The two causes named are defective gyro and frame bearings, and unbalance of the gyro and frame. (L2, PDF p. 633)

187. a Gimbal error of the directional gyro mainly occurs during a turn. (L2, PDF p. 633)

188. b Gimbal error of the directional gyro results in a direction indication error. (L2, PDF p. 633)

189. a In gimbal lock, the extreme case of frame error, the two frames are aligned. (L2, PDF p. 633)

190. a The total effect of topple must be eliminated with an erection system. (L2, PDF p. 635)

191. a The erection system controls the gyro axis so that it keeps a constant direction to the earth's surface. (L2, PDF p. 635)

192. a The reference is detected either by a moving mass such as a pendulum or by liquid level switches. (L2, PDF p. 635)

193. b Depending on the gyro type the erection is done by mechanic force of gravity or by electrical torque motors. (L2, PDF p. 635)

194. a Rotating the apparatus produces a phase displacement between the two counter-running beams. (L2, PDF p. 637)

195. c Optical gyroscopes work according to the optical principle known as the Sagnac effect. (L2, PDF p. 637)

196. c The RLG compares two counterpropagating laser beams in a ring cavity with each other. (L2, PDF p. 637)

197. c Light is coherent if it has been split by reflexion from one single light packet. (L2, PDF p. 641)

198. a Drift is greatest at the poles and belongs to the gyro with its axis parallel to the surface, while topple is zero there. (L3, PDF p. 631)

199. a The axis appears to topple during the journey until it is horizontal at the equator. (L3, PDF p. 633)

200. c Real errors from construction are error values over and above those due to the earth's rotation and transport. (L3, PDF p. 633)

201. c Gimbal lock, where the two frames are aligned, makes the gyroscope unusable. (L3, PDF p. 633)

202. a The erection keeps the axis horizontal for the directional gyro and vertical for the vertical gyro. (L3, PDF p. 635)

203. b The erection is done by mechanic force of gravity or by electrical torque motors depending on the gyro type. (L3, PDF p. 635)

204. b Total topple is the sum of apparent topple and real topple. (L3, PDF p. 636)

205. b In general, the larger the beam cavity is, the easier the interference pattern can be measured. (L3, PDF p. 637)

206. a Beams interfere at extremely small angular velocities, but in such a case they are difficult to measure. (L3, PDF p. 637)

207. c Optical gyroscopes with mirrors are easy to measure but are constrained by the size of the unit. (L3, PDF p. 637)

208. b The beam moving in the direction of rotation travels slightly further than the one running the other way. (L3, PDF p. 639)

209. b The beam on the longer course has the now longer resonator and so a slightly lower frequency. (L3, PDF p. 639)

210. c As long as the wave trains keep the same frequencies the strips remain stationary. (L3, PDF p. 639)

211. a When the frequencies differ the strips wander over the photodiodes, which are alternately in light and dark, and those impulses are metered. (L3, PDF p. 639)

212. a The direction of wander indicates the rotational direction and is taken into account with a preceding symbol when the pulses are added up. (L3, PDF p. 639)

213. b Because coherent light can interfere, speed of rotation, that is arc velocity, can be measured with the structure of the optical circuit. (L3, PDF p. 641)

Laser and fibre optic gyros

PDF pages 643–649 · 19 questions

LEVEL 1 — RECALL

214. To which specification does the fibre optic gyro conform entirely ●

a. ARINC 561
b. ARINC 704
c. ARINC 705

LEVEL 2 — UNDERSTAND AND EXPLAIN

215. Which medium do the gas laser gyroscopes that have prevailed in practice use ●

a. helium/neon
b. neon/krypton
c. helium/argon

216. What determines the frequency of the laser light ●

a. the length of the cavity between cathode and anode
b. the current drawn by the laser exciter
c. the number of mirrors in the light path

217. By which three means can laser rays be produced ●

a. by solid state, liquid or crystal lasers
b. by gas, solid state or semi-conductor lasers
c. by gas, liquid or semi-conductor lasers

218. On what does the sensitivity of a laser gyroscope depend

a. the weight of the gyroscope block
b. the number of legs of the ring cavity
c. the size of the circuit

219. Which component of the fibre optic gyro splits the laser beam into two light waves of equal intensity

a. the light wave conductor spool
b. the photodetector
c. the light beam divider

220. For which aircraft is the fibre optic gyro now used for position and navigation ●

a. short and medium range commercial aircraft, and agricultural aviation
b. short and medium range commercial aircraft, and military aviation
c. long range commercial aircraft, and business aviation

221. The fibre optic gyro reference system is expected to give general aviation an inexpensive and precise position and navigation system when it is combined with ●

a. an air data computer
b. GPS
c. a ring laser gyro

LEVEL 3 — APPLY AND REASON

222. Two laser gyros are built to the same sensitivity but with different ring cavities. Which of the two comes out the smaller unit?

a. the three-leg gyro
b. the four-leg gyro
c. neither, they come out the same size

223. An installation is being changed from a conventional INS platform gyro to a laser gyro. What happens to the wait before the system can be used?

a. it falls away, since the laser gyro has no warm-up and no run-up time
b. it rises, since the laser gyro needs a longer alignment than the platform
c. it is unchanged, since both need the same warm-up before alignment

224. A ring laser gyro has gone into the lock-in range. What has become of its reading?

a. the low rotational speed of the gyroscope block is no longer measurable
b. the gyroscope block reports its highest rotation rate instead
c. the low rotational speed of the gyroscope block is read with doubled accuracy

225. The gyro block is being deliberately shaken back and forth all the time. How is that kept out of the output?

a. the computer compensates for the self-oscillation when calculating the rates
b. the laser exciter is switched off during each back and forth movement
c. the photodiodes are blanked whenever the block moves back and forth

226. Path length control is working on a ring laser gyro. What is moved and to what end?

a. the prisms are corrected to bring the two beams back into phase
b. the photodiodes are corrected to hold the interference strips still
c. the mirrors are corrected to attain the maximum possible brightness

227. A fibre optic gyro is sitting completely still and its two light waves have been brought to interference. What is their relation?

a. they are a quarter wave apart
b. they are exactly opposed
c. they are in phase

228. The fibre optic gyro is now rotated about its vertical axis. Why does a phase displacement appear?

a. the two waves are no longer of equal intensity when they meet
b. the wave running in the direction of rotation is slowed down by the fibre
c. the wave running in the direction of rotation has a longer run-time than the other

229. A phase displacement has built up between the two waves in the coil. What does the photodiode actually see?

a. a break in the interference signal
b. a change in the colour of the interference signal
c. a change in the intensity of the interference signal

230. The input axis of a hemispherical resonator gyro is turned. What does the standing wave do?

a. it grows in amplitude in proportion to the rotational angle
b. it precesses relative to the casing in step with the rotational angle
c. it dies away again until the input axis stops turning

231. Energy is being pumped into the resonator of a hemispherical resonator gyro through the ring forcer electrodes. What is that for?

a. to raise the amplitude of the standing wave with the angle of rotation
b. to restore the damping loss and maintain the resonance frequency
c. to drive the standing wave round against the direction of rotation

232. A fleet planner compares the mean time between failures of the gyro itself with that of the whole inertial reference system. Which of the two is the longer?

a. the system, at 150,000 h against 12,000 h for the gyro
b. they are the same, both being 12,000 h
c. the gyro, at 150,000 h against 12,000 h for the system

Answers — Laser and fibre optic gyros

214. c The fibre optic gyro conforms entirely to the requirements of the ARINC 705 specification. (L1, PDF p. 645)

215. a The gas laser gyroscopes that prevailed use helium/neon as a medium. (L2, PDF p. 643)

216. a The frequency of the laser light is determined by the length of the cavity, cathode to anode. (L2, PDF p. 643)

217. b Laser rays can be produced by gas, solid state or semi-conductor lasers. (L2, PDF p. 643)

218. c The sensitivity of a laser gyroscope depends on the size of the circuit. (L2, PDF p. 643)

219. c The light beam divider is the component that splits the laser beam into the two equal light waves. (L2, PDF p. 645)

220. b The fibre optic gyro is used for position and navigation of smaller short and medium range commercial aircraft and in military aviation. (L2, PDF p. 645)

221. b Together with GPS the fibre optic gyro reference system is intended to offer that system for general aviation. (L2, PDF p. 645)

222. b For the same sensitivity the dimensions of a four-leg laser gyro are smaller than those of a three-leg gyro. (L3, PDF p. 643)

223. a The laser gyro is listed as having neither warm-up time nor gyro run-up time. (L3, PDF p. 643)

224. a In the lock-in range the low rotational speed of the gyroscope block is no longer measurable. (L3, PDF p. 643)

225. a The computer compensates for the self-oscillation when calculating the rotation rates. (L3, PDF p. 643)

226. c Via a photodiode and piezo crystals the mirrors are corrected for maximum brightness and thus the optimal path length. (L3, PDF p. 643)

227. c If there is no rotation the two light waves are in phase at the end of the coil. (L3, PDF p. 645)

228. c Under rotation the beam running with the rotation has a longer run-time and the other a shorter one, which is the Sagnac effect. (L3, PDF p. 645)

229. c The phase displacement leads to a change in the intensity of the interference signal, which the photodiode captures. (L3, PDF p. 645)

230. b When the input axis turns, the standing wave precesses relative to the casing in exact relation to the rotational angle. (L3, PDF p. 647)

231. b Via the ring forcer electrodes the damping loss is restored by pumping energy in to maintain the resonance frequency. (L3, PDF p. 647)

232. c The HRG has an MTBF of 150,000 h and the IRS one of 12,000 h. (L3, PDF p. 649)